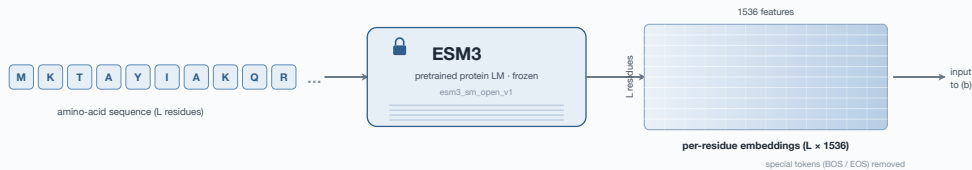
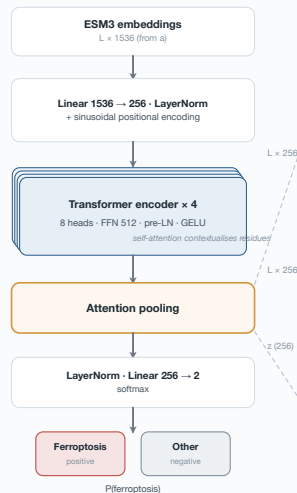


a ESM3 per-residue embedding



b Task classifier



Attention pooling

learned single-query weighting — length-independent & interpretable

$$\text{score } s_i = (q \cdot h_i) / \sqrt{d} \quad a_i = \text{softmax}(s) \text{ over positions } i, \sum a_i = 1$$

